Solution Architecture: The 4-Layer Edge Framework

Project: NeuroClass — Multimodal Attendance & Engagement Agent

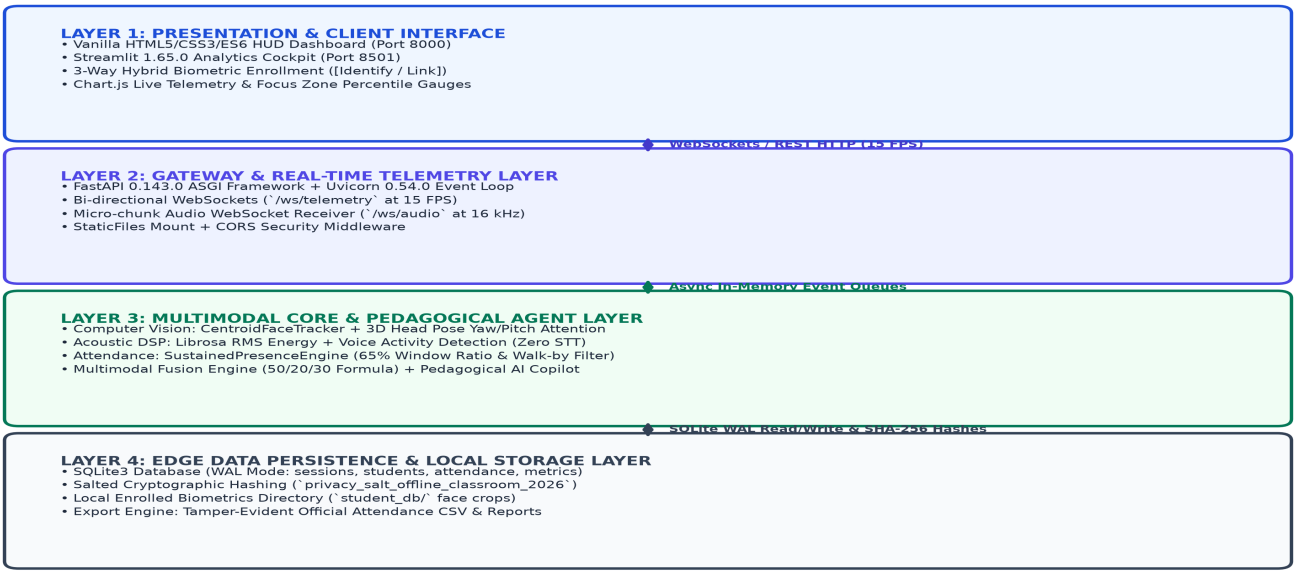
Module: `3. Project Design Phase`

Phase: Technical Architecture & Component Interaction Design

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1. Architectural Overview

NeuroClass is architected as an **Edge-Centric, 4-Layer Multimodal Intelligence Framework**. All computer vision, digital signal processing, fusion heuristics, and database operations execute locally on the host machine.



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2. Layer-by-Layer Architectural Breakdown

The system adheres to a strict separation of concerns across four vertically integrated layers. Data flows sequentially from sensor ingestion through real-time DSP and edge inference, terminating in encrypted local storage and reactive presentation.

Layer 1: Presentation & Client Layer (UI/UX)

- **Vanilla ES6 HUD Interface (`frontend/`):**
- Ultra-lightweight single-page application built without heavy build steps.
- Renders live video HUD with privacy-blurred bounding boxes, track markers, and head pose vectors.
- Displays instant toast notifications for pedagogical nudges.
- Provides the **3-Way Hybrid Enrollment** interface (`[IDENTIFY / LINK] Identify / Link`).
- **Streamlit Analytics Dashboard (`streamlit_app.py`):**
- Companion reporting cockpit providing historical session heatmaps, focus zone distributions, and attendance CSV export.

Layer 2: Gateway & Networking Layer (`backend/api/`)

- **FastAPI Core (`backend/api/server.py`):**
- Manages asynchronous event loop and connection pools.
- Exposes REST routes for session management, student registration, and CSV generation.
- **WebSocket Gateway (`/ws/telemetry` & `/ws/audio`):**
- Delivers low-latency (***<150***ms) bi-directional streaming between the client browser and multimodal backends.
- Ingests microphone audio PCM chunks and pushes processed metrics back to the client.

Layer 3: Multimodal Core & Reasoning Layer (`backend/`)

- **Vision Pipeline (`backend/vision/`, `backend/core/`):**
- Face detection via OpenCV Haar Cascade and MediaPipe mesh.
- Spatial centroid tracking (`CentroidFaceTracker`) preserving track continuity across momentary dropouts.
- Head pose Yaw/Pitch estimation via landmark geometry (`VisualAttentionAnalyzer`).
- **Acoustic Pipeline (`backend/audio/`, `backend/core/audio_analysis.py`):**
- Signal standardization to 16,000 Hz float32 matrices.
- RMS energy dynamics, spectral centroid, spectral rolloff, and zero-crossing rate.
- Voice Activity Detection (VAD) with adaptive energy thresholding.
- **Zero Speech-to-Text (STT) transcription.**
- **Sustained Presence Engine (`backend/attendance/`):**
- Evaluates sliding window presence ratios (***ge 65**\%).
- Filters walk-by noise (***<15***s).
- **Multimodal Fusion & Pedagogical Agent (`backend/fusion/`, `backend/agent/`):**
- Combines visual attention (***50***\%), motion balance (***20***\%), and acoustic rhythm (***30***\%) into composite scores.
- Fires pedagogical interventions with a 45-second anti-fatigue cooldown.

Layer 4: Persistence & Storage Layer (`backend/db/`, `student_db/`)

- **SQLite Database (`classroom_intelligence.db`):**
- Operates in Write-Ahead Logging (WAL) mode for high-concurrency non-blocking reads and writes.
- Stores sessions, students, attendance logs, and engagement time-series.
- Enforces salted hashing (`privacy_salt_offline_classroom_2026`) on all student identifiers.
- **Biometric Directory (`student_db/`):**
- Stores enrolled face reference crops named `{student_id}.jpg`.

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3. Sequence Flow: Live Attendance & Intervention Cycle

